zone between the ‘horizon A’ and ‘horizon C’, and contains matter derived from below as well as from above. It has some organic matter in it, although the mineral matter is noticeably weathered. ‘Horizon C’ is composed of the loose parent material. This layer is the first stage in the soil formation process and eventually forms the above two layers. This arrangement of layers is known as the soil profile.

37. Inceptisols are a soil of an order comprising freely draining soils in which the formation of distinct horizons is not far advanced, such as brown earth. They have no accumulation of clays, iron oxide, aluminium oxide or organic matter.

Ultisols are a highly weathered leached red or reddish-yellow acid soil with a clay-rich B horizon, occurring in warm, humid climates. They are commonly known as **red clay soils**. Ultisols were seen as the ultimate product of continuous weathering of minerals in a humid, temperate climate without new soil formation via glaciation. They are defined as mineral soils which contain no calcareous material anywhere within the soil, have less than weatherable minerals in the extreme top layer of soil, and have less than 35% base saturation throughout the soil.

Urvara and *Usara,* were fertile and sterile soils respectively

38. Saline soils are also known as *Usara* soils. Saline soils contain a larger proportion of sodium, potassium and magnesium, and thus, they are infertile, and do not support any vegetative growth. They have more salts, largely because of dry climate and poor drainage.They lack in nitrogen and calcium.

Due to the dry climate, high temperature and accelerated evaporation, arid soils lack moisture and humus.

The soil develops a reddish colour due to a wide diffusion of iron in crystalline and metamorphic rocks. It looks yellow when it occurs in a hydrated form. The fine-grained red and yellow soils are normally fertile, whereas coarse-grained soils found in dry upland areas are poor in fertility.

39. The lofty Himalayas in the north along with its extensions act as an effective climatic divide. The towering mountain chain provides an invincible shield to protect the subcontinent from the cold northern winds. These cold and chilly winds originate near the